

MAPINN: A Multi-Scale Adaptive Physics-Informed Neural Network for Interface-Dominated and Excitable PDE Systems

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Motivation/ Introduction

Partial Differential Equations (PDEs) are widely used to model complex physical and biological systems such as phase transitions and excitable wave propagation. Physics-Informed Neural Networks (PINNs) solve PDEs by embedding governing equations into neural networks without requiring labeled data. However, conventional PINNs struggle with nonlinear systems involving sharp interfaces and multi-scale dynamics due to spectral bias and inefficient sampling. These limitations often lead to inaccurate solutions and loss of critical physical features.

Scope of the Project

This work aims to develop a robust neural network framework for solving nonlinear PDEs characterized by sharp interfaces and excitable wave dynamics. The study focuses on the Allen-Cahn equation and the FitzHugh-Nagumo system. The objective is to improve prediction accuracy, preserve important physical properties such as interface sharpness and wave propagation, and enhance the learning of multi-scale behavior through an integrated modeling approach.

Methodology

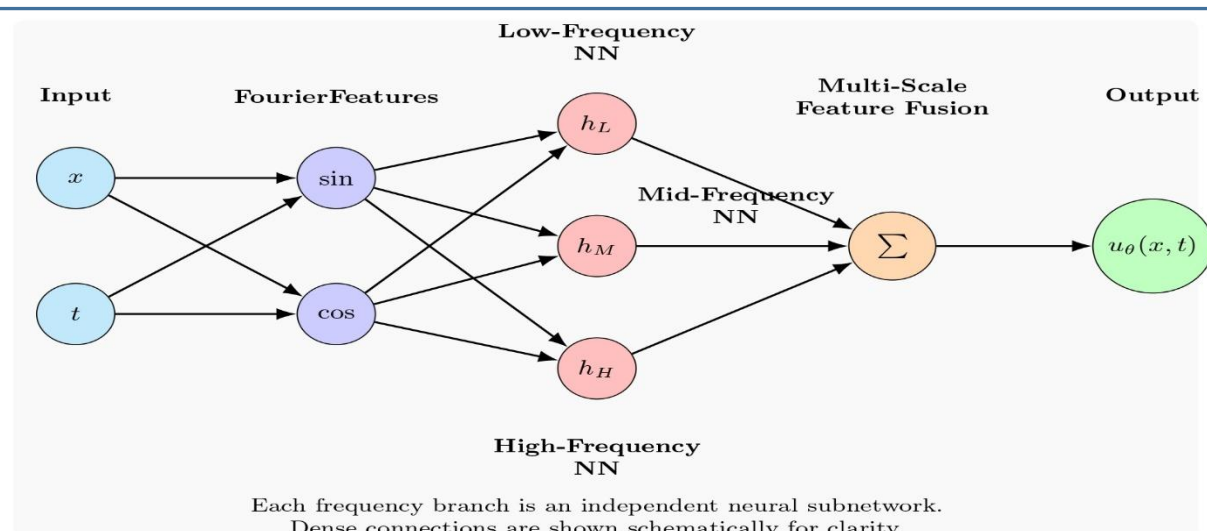


Figure 1: Neural architecture of the proposed MAPINN.

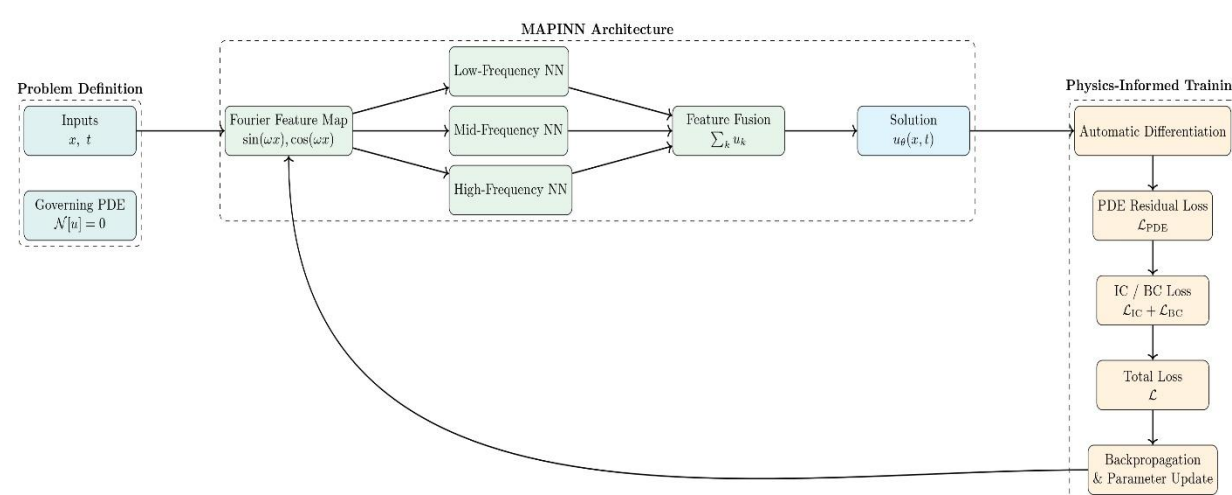


Figure 2: Overall workflow of the proposed MAPINN framework.

MAPINN integrates three core components to address spectral bias and scale entanglement. First, Fourier feature encoding maps input coordinates into a high-dimensional frequency space to capture localized features. Second, a multi-branch architecture processes low, mid, and high-frequency components independently, fusing them to form the final predicted solution:

$$u_{\theta}(x, t) = \sum_{k=1}^K u_{\theta_k}^{((k))}(y_k(x, t))$$

Third, an adaptive residual-based sampling strategy dynamically focuses computational resources on physically active areas like evolving interfaces. The network minimizes a composite physics-informed loss enforcing the initial conditions, boundary conditions, and the governing PDE residuals:

$$L(\theta) = \lambda_f L_{PDE} + \lambda_{ic} L_{IC} + \lambda_{bc} L_{BC}$$

We evaluate this framework on the Allen-Cahn phase-field equation for sharp interface evolution:

$$\frac{\partial u}{\partial t} = \frac{\epsilon^2 (\partial^2 u)}{\partial x^2} - (u^3 - u)$$

We also evaluate it on the FitzHugh-Nagumo excitable system for biological wave propagation:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + u - \frac{u^3}{3} - v \quad ; \quad \frac{\partial v}{\partial t} = \epsilon(u + a - bv)$$

The entire objective is optimized using a two-stage Adam and L-BFGS training strategy to ensure rapid convergence to a high-accuracy physical solution.

Results

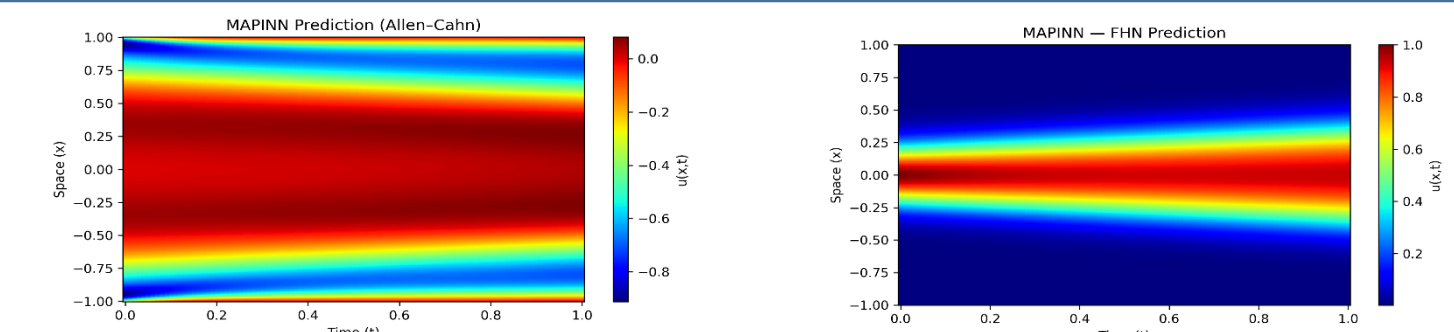


Figure 3

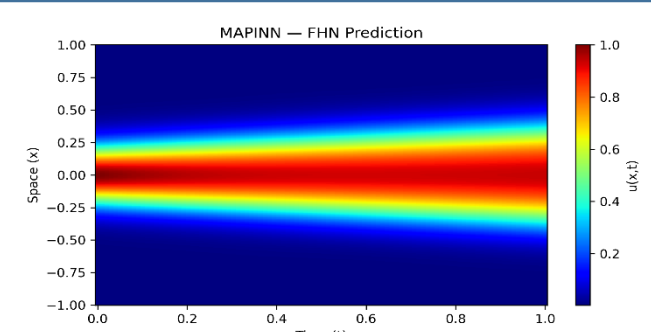


Figure 4

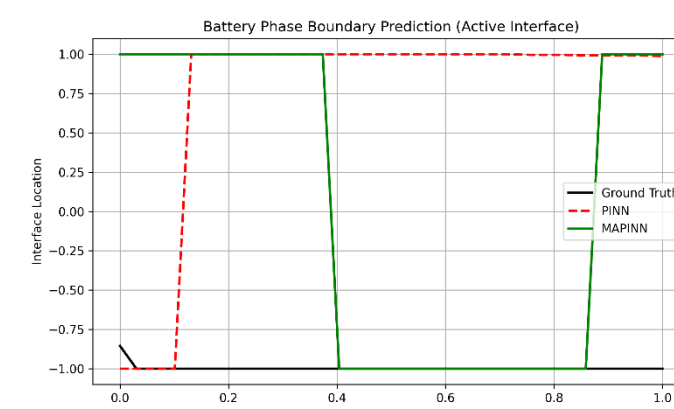


Figure 5

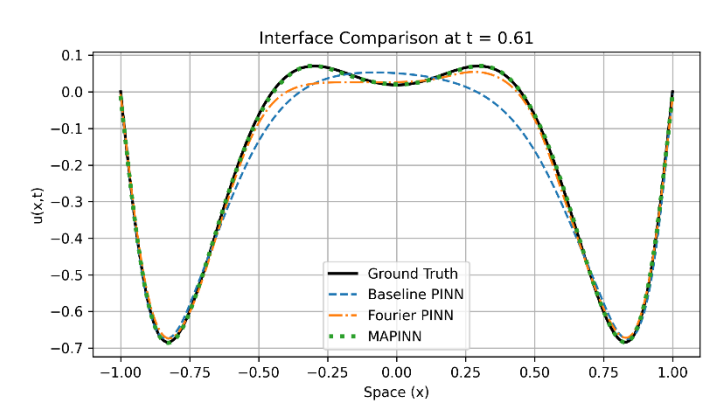


Figure 6

Figure 3: MAPINN Prediction (Allen-Cahn). The solution predicted by MAPINN successfully prevents interface smoothing and resolves sharp gradients, achieving a relative L_2 error of just 2.96% compared to the baseline PINN's 13.47%.

Figure 4: MAPINN prediction of excitation wave. This demonstrates the unique ability of the model to accurately track rapid excitation wavefronts and recovery processes without attenuation in the FitzHugh-Nagumo system, reducing the error to 0.78% (where the baseline PINN failed entirely at 92.13%).

Figure 5: Phase boundary tracking that is based on batteries. MAPINN precisely predicts the active interface locations over time alongside the ground truth, proving its high reliability and viability for real-time digital twin applications.

Figure 6: Interface Comparison at $t = 0.61$. A cross-sectional view proving that MAPINN perfectly preserves sharp interfaces and steep gradients, whereas baseline models suffer from severe interface smoothing.

PDE	Model	Error (%)
Allen-Cahn	Baseline PINN	13.47
	Fourier PINN	5.42
	MAPINN	2.96
FitzHugh-Nagumo	Baseline PINN	92.13
	Fourier PINN	90.03
	MAPINN	0.78

Conclusion/ Summary

Conventional PINNs consistently struggle with spectral bias and interface smearing, but MAPINN overcomes these barriers through a synergistic combination of multi-scale representations and adaptive training. Our results demonstrate that MAPINN reduces global prediction errors by up to two orders of magnitude in excitable dynamics while maintaining high spectral accuracy and physical interface fidelity. Beyond theoretical benchmarks, MAPINN successfully acts as an accurate surrogate model for complex tasks such as battery phase-boundary tracking. This research validates MAPINN as a powerful, mesh-free tool capable of advancing real-time monitoring, predictive control systems, and digital twin technologies across material sciences and biomedical engineering.

Contact Details

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Acknowledgments/ References

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